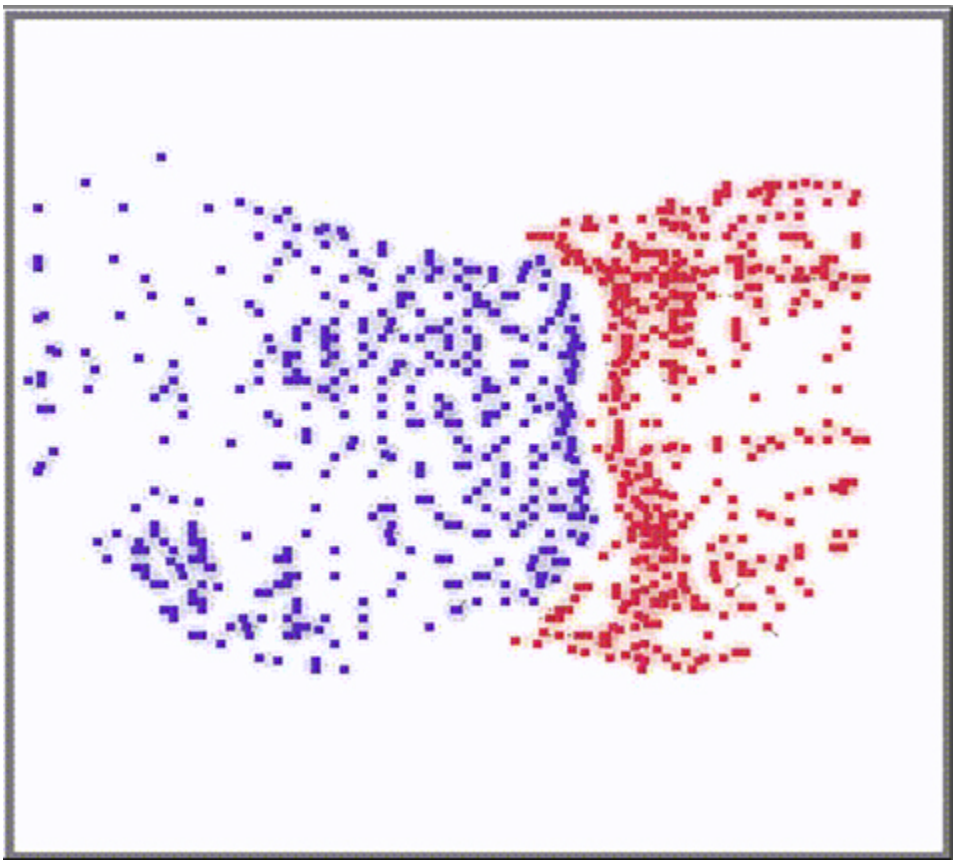


Since all the agents on one team share the same objective and hence the optimal policy, we can use a single DQN to control all the agents on one team, and a different DQN to control the agents on the other team. It's basically a battle between two DQNs, so this would be a perfect opportunity to try out different kinds of neural networks and see which is better. To keep things simple, though, we'll use the same DQN for each team.

You'll need to install the MAgent library from https://github.com/geek-ai/MAgent by following the instructions on the readme page. From this point on, we'll assume you have it installed and that you can successfully run `import magent` in your Python environment.

Listing 9.13. Creating the MAgent environment

```
import magent
import math
from scipy.spatial.distance import cityblock          1

map_size = 30
```